

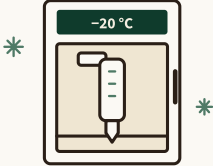
Exhaled Breath Condensate (EBC) Collection

Five steps from the pre-cooled condenser to the analysis laboratory. For nurses and physicians.



Cold chain -20 °C
 unbroken from collection to lab

1



Retrieve the pre-cooled condenser

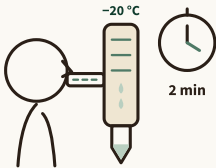
Take the -20 °C breath condenser from the freezer in the sample collection room. Check that the Eppendorf tube is seated at the bottom of the condensation tube and start at once while the tube is cold.

✓ Gloves on. Label the tube: participant ID · date · time.

-20 °C

PRE-COOLED CONDENSER

2



Exhale for 2 minutes

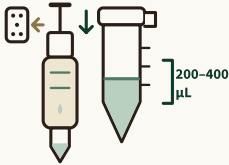
The participant breathes in through the nose and exhales steadily through the mouthpiece into the -20 °C tube for 2 timed minutes. Keep the device upright so the condensate runs down into the Eppendorf tube.

✓ No coughing, talking or saliva into the mouthpiece.

2 min

STEADY EXHALATION

3



Recover the condensed sample

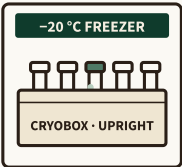
Take out the perforated white piston, then mount the injector pump on the condensation tube. Press it slowly and carefully to drive the condensate into the Eppendorf tube, then detach and cap the tube. Expected yield about 200–400 µL.

✓ Record the volume. Low? Note it; never top up or pool.

200–400 µL

EXPECTED YIELD

4



Store the Eppendorf tube at -20 °C

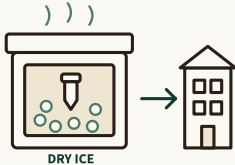
Place the capped, labelled tube upright in the cryobox in the -20 °C freezer immediately after collection and log its position and time.

✓ Once frozen, it stays frozen: no thaw and refreeze.

-20 °C

STORE AT ONCE, UPRIGHT

5



Ship to Gipfel Life Sciences on dry ice

Send frozen samples to the Gipfel Life Sciences laboratory in an insulated box packed with dry ice so they stay at -20 °C or colder throughout. Enclose the sample manifest (ID, date, volume) and notify the lab before dispatch.

✓ Samples must never thaw in transit.

Dry ice

-20 °C OR COLDER IN TRANSIT

CRITICAL POINTS

- ✱ **Cold chain.** Condenser pre-cooled, tube frozen at once, dry ice in transit.
- 🦺 **Clean sample.** No saliva and no coughing into the device.
- 🔗 **Traceability.** Label and log every tube: ID, date, time, volume.

SAMPLE FORM

PARTICIPANT ID	DATE · START TIME
VOLUME, µL	POSITION · INITIALS

Before your breath sample

Food, drinks and smoke change the molecules in your breath for a while, so please follow these simple rules. Sampling is painless and takes about 2 minutes.

COUNTDOWN TO YOUR SAMPLE

2 hours before



Last food. Nothing to eat from now on, not even gum or sweets. No smoking and no sexual intercourse from now on.

1 hour before



Last drink other than water. No coffee, tea, juice, milk, soft drinks or alcohol. Still water is fine at any time.

Sample time



Rinse your mouth with water, sit down, then breathe out steadily through the mouthpiece for 2 minutes.

PLEASE AVOID

- ✗ **Eating** for 2 hours before sampling, including chewing gum, sweets and lozenges.
- ✗ **Any drink except water** for at least 1 hour before sampling: no coffee, tea, juice, milk, soft drinks or alcohol.
- ✗ **Smoking, vaping or nicotine products** for at least 2 hours before sampling, ideally not at all that day.
- ✗ **Mouthwash, throat sprays or lozenges** within 1 hour before sampling.
- ✗ **Sexual intercourse** for at least 2 hours before sampling.
- ✗ **Strenuous exercise** within 1 hour before sampling: a gentle walk is fine.

PLEASE DO

- ✓ **Drink still water** as usual. It does not affect the sample.
- ✓ **Rinse your mouth with water** just before the sample is taken.
- ✓ **Arrive a few minutes early** and rest seated before sampling, so your breathing is calm.
- ✓ **Breathe in through your nose** and out steadily through the mouthpiece for the full 2 minutes.
- ✓ **Take your usual medication** unless your physician has told you otherwise, and tell us what you took.

PLEASE TELL THE TEAM BEFORE SAMPLING IF

- you have had a cold, cough, sore throat or other respiratory infection in the last 2 weeks
- you have asthma, COPD or use an inhaler
- you smoke or vape, or have done so today
- you could not follow one of the rules above: we will note it or find a better time

WHAT HAPPENS DURING SAMPLING

You sit comfortably and breathe out through a clean mouthpiece into a cooled tube for about 2 minutes. Your breath condenses into a few drops of liquid, which are frozen and sent to the laboratory. Nothing is injected and no blood is taken. Questions? Ask the nurse or physician taking your sample.